

Factors for Hyperactivity in Tigrinya

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1 Introduction

(1) *Long-distance hyperagreement with a pseudo-modal predicate*

‘The woman needs to leave.’

(2) *Hyperraising-to-Object of an intensional predicate*

ʔit-a məmħir n=ət-om təmharo₁ [t₁
 that-FSG teacher ACC=that-MPL student.MPL
 ni=ki-xəjd-u] ti-dəlɨj-om
 ACC=IRR-leave.IPFV-SM.3MPL SM.3FSG-want.IPFV-OM.3MPL

‘The teacher wants the students to leave.’

Following a general overview of Tigrinya in §2, I will argue in §3 that these constructions involve two separate derivations in which a single nominal engages in multiple A-relations. We will see that in (1) an embedded argument exceptionally controls object marking on the matrix predicate through a long-distance agreement relationship. In (2), on the other hand, an embedded argument exceptionally raises into the object position of the matrix predicate.

What makes both of these cases exceptional is the fact that the hyperactive nominal is otherwise fully licensed as an argument of the embedded clause. To the extent that this is correct, Tigrinya offers interesting insights into the factors responsible for hyperactive behaviors of nominal constituents. I will argue in §4 that the usual suspects, including Case-licensing and defective domains, which rely on the Activity Condition of Chomsky (2001), do not contribute to an explanatory or predictive account of hyperactivity patterns in Tigrinya.

In §5 I will attempt to motivate an alternative approach that does not rely on the concepts of nominal-licensing or deactivation. The proposal is based on the observation that the hyperactivity patterns of embedded nominals in Tigrinya are predictable on the basis of the argument structure of the matrix predicate. This correlation supports the idea that hyperactivity patterns can be motivated and explained by the properties of verbal functional heads. This idea fits into a larger body of emerging literature that similarly proposes that patterns of hyperactivity reflect formal properties of functional heads on the clausal spine, as opposed to the hyperactive nominals themselves (Zyman 2018; Halpert 2019; Fong 2019; Lohninger et al. 2022; Lee & Yip 2024; Halpert & Zeijlstra 2024). The paper concludes with §6.

2 Background on Tigrinya

2.1 Ethnographic information and data collection

Tigrinya is an Ethiosemitic language on the Semitic branch of the Afro-Asiatic language family. It is, therefore, closely related to the languages Tigré and Amharic and more distantly related to Arabic and Hebrew. Tigrinya is estimated to be spoken by 10 million people. Although the language is spoken primarily in Eritrea and Northern Ethiopia, large immigrant populations exist world-wide (Eberhard et al. 2024). The language is not currently considered endangered.

Unless otherwise noted, the data presented in this paper were collected in individual interviews by the author or collected in collaboration with Gioia Cacchioli. The interviews involved standard judgement tasks that asked the consultant to provide intuitions about the grammaticality or acceptability of utterances in Tigrinya. Test sentences were typically paired with contexts or translations to demonstrate the intended interpretation and to facilitate judgments. The primary consultant for the research reported here, who is proficient in English, is from the Tigray region of Northern Ethiopia and currently lives in Addis Abeba. The other three consultants are from the Gash-Barka and Debub regions of Eritrea and are proficient in French. With the notable exceptions of Tsehaye (2009) and Niguss (2021), the non-trivial variation between dialects of Tigrinya remains under-explored.

2.2 Basic morphosyntax

As is typical of Semitic languages, tri-consonantal roots in Tigrinya acquire their category and certain inflectional information through transfixes and vowel templates (Buckley 2003; Godfrey 2011). This is the case for the different aspectual verb forms of the paradigmatic root /sbr/ “break” adapted from Leslau 1941.¹

- (3) a. *Imperfective*
ji-səbbir
SM.3MSG-break.IPFV
‘It breaks.’

¹As Nazareth 2011 discusses, one or more of the traditional names for the aspectual verb forms in Tigrinya are possibly misnomers. For instance, the “gerundive” verb form typically expresses perfective aspect in matrix clauses, although its name reflects its wider distribution in the language.

- b. *Gerundive*
səbir-u
break.GER-SM.3MSG
'It broke.'
- c. *Perfect*
səbər-ə
break.PRF-SM.3MSG
'It has broken.'

In addition to subject and object agreement markers, verbs also host prefixal morphology that indicates finiteness, mood, valency alternations, and subordination relations.²

Although atypical for Semitic languages, Tigrinya is a fairly well-behaved SOV language with a strongly head-final verbal domain. Illustrative examples with simplex sentences are provided below in (4).

(4) *Causative-inchoative alternation in Tigrinya*

- a. jonas n=ət-a t'irmuz səbir-u-wa
Yonas.M ACC=that-FSG bottle.F break.GER-SM.3MSG-OM.3FSG
'Yonas broke the bottle.'
- b. ʔit-a t'irmuz tə-səbir-a
that-FSG bottle INTR-break.GER-SM.3FSG
'The bottle broke.'

(Nazareth 2011: 56, (55))

As the examples above also serve to illustrate, Tigrinya is a nominative-accusative language with respect to both case-marking and agreement alignment. The grammatical subject of transitive and intransitive predicates is morphologically unmarked with nominative case and controls subject marking (SM) morphology on the predicate. Grammatical objects are differentially marked with the accusative case preclitic *n(i)=* on the basis of factors like definiteness and specificity while other objects are unmarked (Nazareth 2011). It is typical for accusative-marked nominals in mono-clausal structures to both control object marking (OM) morphology on the verb and undergo Object Shift (Overfelt 2022).³

²Prefixal subject markers with initial glide and laryngeal consonants are often deleted after other prefixes. For ease of exposition, I will typically omit this information from interlinear glosses. Full agreement paradigms can be found on pages xvi–xvi.

³While object marking and accusative case marking are strongly correlated, both of these are

3 Hyperactivity in Tigrinya

The empirical domain of the paper involves the *ki*-clause complements of pseudo-modal predicates in (5) and intentional predicates in (6). These clauses are so-called for the prefix *ki*-, which is identified here as an irrealis marker and appears on the embedded predicate.⁴ Of interest is the fact that a nominal constituent in each case controls agreement marking on both the embedded and matrix predicate.

(5) *Embedded ki-clause complement of a pseudo-modal predicate*

ʔit-a səbajti ki-t-xəjjid ji-gibaʔ-(a)
that-FSG woman.F IRR-SM.3FSG-leave.IPFV SM.3MSG-need.IPFV-OM.3FSG

‘The woman needs to leave.’

(6) *Embedded ki-clause complement of an intensional predicate*

ʔit-a məmh̄ir n=ət-om təmharo
that-FSG teacher ACC=that-MPL student.MPL
ni=ki-xəjd-u ti-dəlij-om
ACC=IRR-SM.3MPL-leave.IPFV-SM.3MPL SM.3FSG-want.IPFV-OM.3MPL

‘The teacher wants the students to leave.’

The following subsections will argue that, in both constructions, an embedded nominal hyperactively engages in multiple A-relations with different predicates. However, we will also see that this hyperactive behavior manifests differently. The *ki*-clause complement of pseudo-modal predicates are transparent for long-distance AGREE (§3.1) while the *ki*-clause complement of intensional predicates are transparent for Raising (§3.2).

3.1 Hyperagreement with pseudo-modal predicates

Pseudo-modal predicates and their *ki*-clause complements, such as /gbʔ/ ‘need’ in (7), have been a focus of attention for Tesfay (2016), Gebregziabher (2021), Cacchioli & Overfelt (in preparation), and Overfelt & Cacchioli (under review).

tied to notions like definiteness, specificity, epistemic familiarity, discourse prominence, and topicality. See Nazareth 2011 and Gebregziabher 2021 for discussion.

⁴See Cacchioli (in preparation) for discussion of the morpheme *ki* and its distribution.

(7) *Analysis of pseudo-modal predicates with ki-clause complements*

expl [ʔit-a səbajti ki-t-xəjjid]
that-FSG woman.F IRR-SM.3FSG-leave.IPFV
ji-giba?-(a)
SM.3MSG-need.IPFV-OM.3FSG

‘The woman needs to leave.’

While analyses vary, [Cacchioli & Overfelt \(in preparation\)](#) and [Overfelt & Cacchioli \(under review\)](#) propose the representations provided above. The logical subject *ʔita səbajti* ‘the woman’ is generated as an argument of the embedded clause, where it receives structural Case and triggers agreement on the embedded predicate. Object marking on the matrix predicate is argued to be the reflex of a formal AGREE relationship with the matrix v^0 and is fed by an application of scrambling to the edge of the embedded clause (e.g., [Polinsky & Potsdam 2001](#)). The default 3MSG subject marker on the pseudo-modal predicates indicates the absence of a referential grammatical subject in the matrix clause in addition to the unavailability of a Raising-to-Subject derivation.

Positional restrictions on the unmarked bare nominal expression *səb* ‘person, people’ offer one indicator that hyperactive nominals in pseudo-modal constructions are generated within the embedded clause. Observe in (8) that *səb* can only be interpreted in subject positions; licensing the use of *səb* in object positions requires the accusative case-marker.

- (8) a. *səb pro* *ħagiz-u-ni*
 person 1SG help.GER-SM.3MSG-OM.1SG
 ‘Someone helped me.’
 b. *pro* **(ni=)səb* *ħagiz-e*
 1SG ACC=person help.GER-SM.1SG
 ‘I helped someone.’

This makes it possible to diagnose a grammatical subject position by testing for the grammaticality of *səb* in that position. With this in mind, observe that *səb* grammatically serves as the logical subject in a pseudo-modal construction:

- (9) *expl* [**səb** *ni=gəz-u* *ki-xəjjid*]
 person LOC=house-POSS.3MSG IRR-SM.3MSG.leave.IPFV
ji-gibo?-o
 SM.3MSG-need.IPFV-OM.3MSG
 ‘Someone needs to go to his house.’

This suggests that, despite controlling matrix object marking, the hyperactive nominal is not generated as an object of the matrix clause. Instead, it appears to serve as a grammatical subject of the embedded clause.

A second indicator of the embedded origin of hyperactive nominals is found in their relative ordering with adverbials. In example (10) below, the adverbial modifier *bibikeri* ‘with a glass’ of the embedded predicate can precede the hyperactive nominal *Segen*.

- (10) *expl* [{*bi=bikeri*} **səgen** {*bi=bikeri*} *maj* *ki-ti-sətti*]
 INS-cup Segen.F INS-cup water IRR-SM.3MSG-drink.IPFV
ji-giba?-a
 SM.3MSG-need.IPFV-OM.3MSG
 ‘Segen needs to drink water with a cup.’

This observation suggests that *Segen* is a constituent of the matrix clause, which is made more compelling by data like (11). The inability for *?ab dʒardin* ‘in the garden’ to appear clause-initially, before the grammatical subject of the matrix intensional predicate, reflects the dispreference in Tigrinya for the long-distance scrambling of adjuncts.⁵

- (11) {**?ab dʒardin*} *təsfaɟ* [{*?ab dʒardin*} *?ane ki-həgiz-o*]
 LOC garden Tesfay.M LOC garden I IRR-help-OM.3MSG
ji-ts’əbijj-ni
 SM.3MSG-expect.IPFV-SM.1SG
 ‘Təsfaɟ expects me to help him in the garden.’

Knowing that adjuncts resist long-distance scrambling, the ability for *bibikeri* ‘with a cup’ to appear utterance-initially in (10) suggests that it, and therefore *Segen*, are constituents of the embedded clause.

⁵Spadine (2020) observes that arguments also resist long-distance scrambling. But personal research suggests that this holds only for complement clauses with more highly articulated structures, including the *kəmzi*-clauses investigated in Spadine 2020 and in .

A second aspect of the analysis in *Overfelt & Cacchioli under review* is that the exceptional object marking on the matrix predicate is fed by scrambling to the edge of the embedded clause. This is opposition to the Hyperraising-to-Object analysis in the following subsection. The example in (9) is consistent with this claim. The hyperactive nominal behaves like a constituent of the embedded clause while still controlling matrix object marking.

Morphological case marking also suggests that the hyperactive nominal remains in the embedded clause. If there were a derived object position in the matrix to receive the hyperactive nominal, the expectation should be that it could be marked with accusative case, contrary to fact:

- (12) * **n=ət-a** **səbajti** **ki-ti-xəjjid** **ji-gibba?-(a)**
ACC=that-FSG woman IRR-S.3fs-leave.IPFV SM.3MSG-need.IPFV-OM.3FSG
Intended: ‘The woman needs to leave.’

Instead, a hyperactive nominal in a pseudo-modal construction is always marked with the case that is appropriate for its grammatical role in the embedded clause. This can be seen easily in sets of examples like those in (13) provided below. Given both a subject and object argument embedded under a pseudo-modal predicate, either of them is principle capable of controlling object marking in the matrix clause. Moreover, each nominal is appropriately case-marked on the basis of its grammatical role in the embedded clause.

- (13) a. [**?it-a** **səbajti** **n=ət-om** **məts’haf-ti**
that-FSG woman ACC=that-MPL book-PL
ki-ti-nbib-om] **ji-giba?-a**
IRR-SM.3FSG-read.IPFV-OM.3MPL SM.3MSG-need.IPFV-OM.3FSG
‘The woman needs to read the book.’
b. [**?it-a** **səbajti** **n=ət-om** **məts’haf-ti**
that-FSG woman ACC=that-MPL book-PL
ki-ti-nbib-om] **ji-giba?-om**
IRR-SM.3FSG-read.IPFV-OM.3MPL SM.3MSG-need.IPFV-OM.3MPL
‘The woman needs to read the book.’

Examples of this type are also informative with respect to the operation that feeds the exceptional object marking. The descriptive lack of Minimality effects, which should otherwise be expected under a Raising-to-Object analysis, point toward an alternative account. On this basis of these and other facts,

Overfelt & Cacchioli (under review) conclude that A'-scrambling, which may be covert in instances such as (13b), places nominal constituents in a sufficiently local relationship with the matrix v^0 to feed object agreement.

3.2 Hyperraising-to-Object of intensional predicates

Intensional predicates, such as the root /ts'bj/ 'expect', shown above in example (10), and /dlj/ 'want' below, have received relatively less attention in the theoretical literature on Tigrinya.

(14) *Analysis of intensional predicates with ki-clause complements*

?it-a məmħir n=ət-om təmharo₁ [t₁
 that-FSG teacher ACC=that-MPL student.MPL
 ni=ki-xəjd-u] ti-dəlij-om
 ACC=IRR-leave.IPFV-SM.3MPL SM.3FSG-want.IPFV-OM.3MPL

'The teacher wants the students to leave.'

Overfelt & Cacchioli (under review) conclude that these examples are similarly derived by way of a scrambling-fed long-distance hyperagreement relationship. On the basis of new data, I will argue here that these constructions indeed are derived via Hyperraising-to-Object, which is both optional and possibly covert (Deal 2017; Zyman 2018; Fong 2019).

The proposed derivation is presented in (14), wherein the logical subject of the embedded clause ?itom təmharo 'those students' is generated as an argument of the embedded predicate. It is within the embedded *ki*-clause that this argument triggers expected agreement on the embedded predicate and, by hypothesis, receives structural Case. As an instance of Raising, object agreement on the matrix predicate is fed by movement of the structurally highest argument to a derived object position in the matrix clause. Like typical Object Shift in the language, this correlates with the valuation of the ϕ -features determining the shape of the object marker on the predicate.

To the extent that the distribution of *səb* can diagnose grammatical subjects, examples like (15) suggests that the logical subject of the embedded predicate can serve as the grammatical subject of the embedded clause, particularly in the case that it does not control matrix object marking.

- (15) *ʔit-a səbajti [səb ki-Ø-xəjjid] ti-dəlij*
 that-FSG woman person IRR-SM.3MSG-leave.IPFV SM.3FSG-want.IPFV
 ‘The woman wants people to leave.’

We have already seen that, with respect to their ordering with modifiers, hyperactive nominals in constructions with intensional predicates behave like constituents of the embedded clause; recall (11). In example (16) as well, the hyperactive nominal *ʔitom təmharo* ‘those students’ follows an adjunct modifier of the embedded predicate.

- (16) *ʔit-i məmhīr [{ʔbiq’ilt’uf} ʔit-om təmharo {biq’ilt’uf}*
 that-MSG teacher quickly that-MPL student.PL quickly
ki-xəjd-u] ji-dəlj-om
 IRR-leave.IPFV-SM.3MPL SM.3MSG-want.IPFV-OM.3MPL
 ‘The teacher wants the students to quickly leave.’

Although it is the marked option, the hyperactive nominal in (16) above optionally appears in the embedded clause and, notably, is unmarked for nominative case while controlling matrix object marking.⁶ The unmarked construction in which the embedded nominal controls object marking in the matrix clause is shown in (17). This example reflects the preference for matrix object marking to be accompanied by accusative case marking on the hyperactive nominal.

- (17) *ʔit-i məmhīr n=ət-om təmharo₁ [t₁*
 that-FSG teacher ACC=that-MPL students
ni=ki-xəjd-u] ji-dəlj-om
 ACC=IRR-leave.IPFV-SM.3MPL SM.3FSG-want.IPFV-OM.3MPL
 ‘The teacher wants the students to leave.’

The presence of accusative case suggests that matrix object agreement is associated with a nominal that shifts to a derived object position, as postulated by a Hyperraising-to-Object derivation.

We should, therefore, expect that a hyperactive nominal would show other behaviors characteristic of occupying a grammatical object position. The example in (18) shows that, like true objects, the logical subject of the embedded predicate can be promoted to subject when the intensional predicate is passivized.

⁶It is observations of this type that contribute to the claim that Hyperraising-to-Object here may not always be overtly realized in the surface string.

- (18) *?it-om tāmharo*₁ b=it-i māmhir [*t*₁
 that-MPL student.PL INS=that-MSG teacher
 ni=ki-xəjd-u] tə-dəlɨj-om
 ACC=IRR-leave.IPFV-SM.3MPL INTR-want.GER-SM.3MPL
 ‘The students are wanted by the teacher to leave.’

In the example above, the logical subject of the embedded predicate *?itom tāmharo* ‘those students’ now controls subject agreement on the matrix predicate and surfaces to the left of the demoted agent *biti memhir* ‘by the teacher’. These are properties that one would expect from promotion to the grammatical subject position of the matrix clause. This possibility suggests that *?itom tāmharo* at some point occupies the matrix object position.

Finally, we find evidence for a difference in the mechanism responsible for feeding object marking on an intensional predicate. The examples in (19) demonstrate the presence of Minimality effects. Descriptively, matrix object marking only cross-references the structurally highest embedded argument, namely the subject.

- (19) a. *?it-i māmhir n=et-om tāmharo* [*t*₁ n=ət-a məts’haf
 that-MSG teacher that-MPL student.PL ACC=that-FSG book
 ki-nbib-u-wa] ji-dəlɨj-om
 IRR-read.IPFV-SM.3MPL-OM.3FSG SM.3MSG-want.IPFV-OM.3MPL
 ‘The teacher wants the students to read the book.’
 b. **?it-i māmhir* [*?it-om tāmharo n=ət-a məts’haf*
 that-MSG teacher that-MPL student.PL ACC=that-FSG book
 ki-nbib-u-wa] ji-dəlɨj-a
 IRR-read.IPFV-SM.3MPL-OM.3FSG SM.3MSG-want.IPFV-OM.3FSG
 ‘The teacher wants the students to read the book.’

These Minimality effects are expected given the typical A-movement operation that is supposed to generate Raising configurations.

4 Non-predictive and non-explanatory factors for hyperactivity

The previous section argued that nominals under the pseudo-modal predicate */gbʔ/* ‘need’ engage in hyperagreement with the matrix predicate while a nomi-

nal embedded under the intensional predicate /dlj/ ‘want’ undergoes hyperraising into a derived matrix object position. The theoretical questions that arise at this point are what permits hyperactivity in Tigrinya and what determines the different hyperactivity patterns.

There is a history of accounting for patterns of (non-)hyperactivity that is built upon the idea that nominal constituents must be licensed by formal syntactic features, notably including Case (Vergnaud 1977/2008, et seq.). An influential implementation of this idea, which has its origin in Chomsky 2001, holds that nominal constituents become inaccessible to additional agreement relations and A-movement upon being licensed, as per the condition in (20):

- (20) *Generalized Activity Condition* (adapted from Chomsky 2001: 6)
A nominal constituent that is formally licensed is inactive.
An inactive nominal constituent is not accessible to A-relations.

This approach opens the door for hypothesizing that certain syntactic domains, which are deficient on some measure, are defective, meaning that nominals may fail to be licensed. Other approaches propose that Case may not be a relevant feature for licensing and deactivating nominals.⁷

I will argue that accounts of (non-)hyperactivity that rely on nominal-licensing or concepts of defectiveness do not provide an explanation for the patterns of hyperactivity in Tigrinya. Moreover, Tigrinya offers compelling evidence that such concepts fail to predict patterns of hyperactivity cross-linguistically. These observations will motivate the conclusion that the Activity Condition does not productively contribute to an understanding of hyperactivity in Tigrinya and beyond (e.g., Bhatt 2005; Nevins 2005; see also Baker 2008).

4.1 Against deactivation by Case-licensing

The prevalence of hyperactivity in Bantu languages, coupled with a persistent lack of evidence for Case-licensing, has given rise to the hypothesis that Case is not a relevant factor in the licensing of nominal constituents, at least in a particular set of those languages (Carstens 2011; Diercks 2012; Carstens & Diercks 2013; see also Van der Wal 2015; Sheehan & van der Wal 2018). Supposing that the same were true for Tigrinya, hyperactivity patterns could be understood as the reflex of the absence of a deactivating feature for nominals.

⁷See Zyman 2023 and Fong & Halpert to appear for overviews of the empirical and theoretical landscape of hyperactivity.

This is a position that Overfelt & Cacchioli (under review) have previously argued against. As they note, Tigrinya is like other Ethiosemitic and Semitic languages in showing several indicators of being a Case-licensing language (see especially Weldu 2004 and Nazareth 2011). There are environments in which nominal constituents must be marked with specific morphology, undergo displacement, or both. These are precisely the kinds of morphosyntactic behaviors that traditional Case-Theory is intended to explain (Vergnaud 1977/2008; Chomsky 1981; see Sheehan & van der Wal 2018 for discussion).

The behavior of definite/specific direct objects, such as *nəta dəbdabe* ‘that book’ in (21a), presents a relevant example. Such arguments are typically morphologically marked with the pre-clitic *n(i)=*, which is commonly analyzed as the exponence of accusative and dative Case (Weldu 2004; Kievit & Kievit 2009; Nazareth 2011; Gebregziabher 2013). Direct objects carrying this morpheme also typically undergo Object Shift to appear to the left of adverbials such as *k’olt’ifu* ‘quickly’. This is unlike the unmarked indefinite counterpart in (21b), which typically remains adjacent to the verb.

- (21) a. *ʔit-i təmaharaj n=ət-a dəbdabe k’olt’if-u*
that-MSG student ACC=that-FSG book quickly-3MSG
ts’ihif-u-wa
read.GER-SM.3MSG-OM.3FSG
‘The student quickly wrote the letter.’
b. *ʔit-i təmaharaj k’olt’if-u dəbdabe ts’ihif-u*
that-MSG student quickly-3MSG letter wrote.GER-SM.3MSG
‘The student quickly wrote a letter.’

The behavior of nominal arguments under passivization also points toward the requirement for traditional Case-licensing. The example in (22) presents a passive version of example (21a).

- (22) *ʔit-a dəbdabe b=it-i təmaharaj tə-ts’ahif-a*
that-FSG letter INS=that-MSG student INTR-write.GER-SM.3FSG
‘The letter was written by the student.’

The theme argument in the passive construction above has been promoted to the grammatical subject position, ensuring that it appears clause initially, controls subject marking on the predicate, and is unmarked for nominative Case. Additionally, the optional agent argument appears marked with the preposition *b(i)=*

and follows the promoted theme.

Now consider again the Tigrinya examples repeated below. They contain hyperactive nominals that are constituents of the embedded clause, as indicated by their linear positioning. These nominals must, therefore, be Case-licensed within the embedded clause. Indeed, in (23b), the hyperactive nominal is overtly marked with accusative case.

- (23) a. ?ʔit-i məmhɪr [biq'ilt'uf ʔit-om təmharo
that-MS teacher quickly that-MP student.PL
ki-xəjd-u] ji-dəlj-om
IRR-leave.IPFV-S.3MP S.3MS-want.IPFV-O.3MP
'The teacher wants the students to quickly leave.'
- b. [ʔit-a səbajti n=ət-om mətə'shaf-ti
that-FSG woman ACC=that-MPL book-PL
ki-ti-nbib-om] ji-giba?-om
IRR-SM.3FSG-read.IPFV-OM.3MPL SM.3MSG-need.IPFV-OM.3MPL
'The woman needs to read the book.'

This suggests that it is not possible to develop an explanatory theory of hyperactivity in Tigrinya purely by appealing to Case and Case-licensing. Nominal constituents that are clearly licensed and Case-marked within the embedded clause remain accessible to additional A-relations. Moreover, hyperactivity is not a Case discriminating phenomenon in the language. Both unmarked nominative arguments and marked accusative arguments are targets for multiple A-relations. It is also not clear that Case-licensing could offer an explanation for how hyperactivity manifests given that we are observing hyperagreement under pseudo-modal predicates and hyperraising from under intensional predicates.

It does not appear that Case and Case-licensing contribute to a predictive theory of which languages display nominal hyperactivity either. We observe hyperactivity in Tigrinya despite it behaving like a Case-licensing language. Thus, being a Case-licensing language does not determine that a language will resist hyperactive behaviors (see also Baker 2008). These results add to previous conclusions that should push us away from theories of hyperactivity that rely on Case or Case-licensing (see Carstens & Diercks 2013 and Bantu languages and Keine 2018 on Hindi-Urdu).

4.2 Against defective domains and non-deactivation

Building on Chomsky (1981; 2000), a more traditional line of analyses proposes that the accessibility of a nominal constituent for A-relations is predictable on the basis of the defectiveness of syntactic domains. As stated in (24), heads that might otherwise license and deactivate nominal constituents may be defective by virtue of being featurally incomplete.

(24) *Defectivity Condition*

Featurally incomplete heads are defective.

Defective heads cannot license nominal constituents.

The correlation between featural deficiency and nominal activity motivates the assertion that, among the other defective properties of such heads, is the inability to license, and thereby deactivate, nominal constituents. We will look at two such proposals only to find that they neither explains nor predicts hyperactivity patterns in Tigrinya.

4.2.1 Partial agreement and defectiveness

Defectiveness may be reflected in instances of deficient specification for ϕ -features, including person, number, and gender. This is the explanation that Chomsky (2000) offers for what Carstens (2001) refers to as “multiple agreement” in the French example below:

(25) *Multiple agreement in French*

Elle est mort-e
she be.3SG dead-FSG

‘She is dead.’

(Carstens 2011: 148, (1))

Observing that partial agreement for only person and number with the adjectival predicate *morte* ‘dead’ does not preclude *elle* ‘she’ from agreeing with the auxiliary, it is hypothesized that ϕ -incomplete agreement does not deactivate nominals.

Nunes (2008) and Ferreira (2009) offer analyses in the same spirit for hyper-raising in Brazilian Portuguese, as in (26) below.

(26) *Hyperraising in Brazilian Portuguese*

o João₁ disse [que *t*₁ **comprou** um carro]
 the João said.3SG that bought.3SG a car

‘João said that he bought a car.’ (Nunes 2008: 87, (4))

The embedded predicate *comprou* ‘bought’ partially agrees with *o João* in only person and number. This motivates the hypothesis that the T⁰ of the embedded clause in the example above is ϕ -incomplete. It is, therefore, defective and leaves *o João* accessible to A-relations with the matrix predicate.

Let us consider the agreement paradigms for the subject and object marking morphemes in Tigrinya that are presented respectively in Table 1 and Table 2. Both the subject and object marking paradigms show (almost) entirely non-syncretic morphological forms for each combination of ϕ -features within any given aspectual verb form.⁸

Table 1: Subject marker paradigms by aspectual verb form in Tigrinya

Subject	Imperfective	Gerundive	Perfect
SM.3MSG	ji -säbbir	säbir- u	säbär-ä
SM.3MPL	ji -säbir- u	säbir- om	säbär- u
SM.3FSG	ti -säbbir	säbir- a	säbär-ät
SM.3FPL	ji -säbir- a	säbir- än	säbär- a
SM.2MSG	ti -säbbir	säbir- ka	säbär- ka
SM.2MPL	ti -säbir- u	säbir- kum	säbär- kum
SM.3FSG	ti -säbir- i	säbir- ki	säbär- ki
SM.3FPL	ti -säbir- a	säbir- kin	säbär- kin
SM.1SG	?i -säbbir	säbir- ä	säbär- ku
SM.1PL	ni -säbbir	säbir- na	säbär- na

⁸The one instance of syncretism is found in Table 1 between the 3FSG and 2MSG subject marker in the imperfective.

Table 2: Object marker paradigms by aspectual verb form in Tigrinya

Object	Imperfective	Gerundive	Perfect
OM.3MSG	ji-k'ätl-o	k'ätil-u-wo	k'ätäl-Ø-o
OM.3MPL	ji-k'ätl-om	k'ätil-u-wom	k'ätäl-Ø-om
OM.3FSG	ji-k'ätl-a	k'ätil-u-wa	k'ätäl-Ø-a
OM.3FPL	ji-k'ätl-än	k'ätil-u-wän	k'ätäl-Ø-än
OM.2MSG	ji-k'ätlä-kka	k'ätil-u-kka	k'ätäl-ä-kka
OM.2MPL	ji-k'ätlä-kkum	k'ätil-u-kkum	k'ätäl-ä-kkum
OM.2FSG	ji-k'ätlä-kki	k'ätil-u-kki	k'ätäl-ä-kki
OM.2FPL	ji-k'ätlä-kkin	k'ätil-u-kkin	k'ätäl-ä-kkin
OM.1SG	ji-k'ätlä-nni	k'ätil-u-nni	k'ätäl-ä-nni
OM.1PL	ji-k'ätlä-nna	k'ätil-u-nna	k'ätäl-ä-nna

In cases of hyperactivity in Tigrinya, provided again below, an argument of the embedded clause controls non-syncretic, ϕ -complete agreement on both the embedded predicate and the matrix predicate.

- (27) a. *expl* [**?it-om** **təmhəro** ki-xəjd-u]
that-MPL student.PL IRR-leave.IPFV-SM.3MPL
ji-giboʔ-om
SM.3MSG-need.IPFV-OM.3FSG
‘The students need to leave.’
- b. **?it-i** **məmhir n=ət-om** **təmhəro**₁ [*t*₁
that-FSG teacher ACC=that-MPL students
ni=ki-xəjd-u] **ji-dəlj-om**
ACC-IRR-leave.IPFV-SM.3MPL SM.3FSG-want.IPFV-OM.3MPL
‘The teacher wants the students to leave.’

It is not clear, therefore, that a theory of defectiveness that is built on the completeness of ϕ -features can provide a non-stipulative account for hyperactivity in Tigrinya. There is no obvious subset of ϕ -features expressed in the language that can plausibly be said to be absent from the embedded clause to result in deficiency. Moreover, given that ϕ -complete agreement is observed both constructions, it is not clear that agreement offers a relevant factor in explaining the nature of the hyperagreement pattern observed in each instance.

It is also unclear that ϕ -feature incompleteness offers a predictive account

of hyperactivity cross-linguistically. While hyperactivity arises in French and Brazilian Portuguese due to purported ϕ -incomplete agreement, hyperactivity in Tigrinya arises even in the context of ϕ -complete agreement. Thus, Tigrinya adds to previous results that speak against theories of (non-)hyperactivity that rely on the ϕ -completeness of licensing heads (see also Carstens 2001; 2011 on Bantu languages).

4.2.2 Tenselessness and defectiveness

Defectiveness has also been claimed to be a property of an embedded T^0 that is not specified for semantic tense. Alexiadou & Anagnostopoulou (1999) offer this account for instances of Greek hyperraising below.

(28) *Hyperraising in Greek*

Ta pedhia arxisan [na trexoun (*avrio)]
the children started.3PL SBJV run.3PL tomorrow

lit. ‘The children started [to run (*tomorrow)].’

(Zyman 2023: 33–34, (10)–(11))

The nominal constituent *ta pedhia* ‘the children’ controls agreement in the embedded clause while at the same time Raising to the matrix subject position where it controls agreement in the matrix clause. The relevant observation is that these properties emerge in the context of embedded subjunctive clauses. Such clauses notably do not permit tense adverbs that specify a time different from the matrix time of evaluation, suggesting that the embedded clause is not specified for semantic tense. By hypothesis, this defect prevents the embedded T^0 from licensing and deactivating *ta pedhia*, thereby leaving it accessible to additional A-relations with the matrix clause.

Cacchioli & Overfelt (in preparation) argue that complement *ki*-clauses in Tigrinya are structurally reduced. This offers an explanation for the fact that these clauses cannot host tense-expressing auxiliaries:

- (29) a. ?it-a səbajti ti-nəbīb (*ki-ʔall-a)
that-FSG woman.F SM.3FSG-read.IPFV IRR-AUX.NPST-SM.3FSG
ji-gibaʔ-a
SM.3MSG-need.IPFV-OM.3FSG
Intended : ‘The woman needs to be reading.’

- b. ʔit-a məmhīr n=ət-om təmhāro
 that-FSG teacher ACC=that-MPL student.PL
 ji-nbīb-u (*ki-ʔall-om)
 SM.3MPL-read.IPFV-SM.3MPL IRR-AUX.NPST-SM.3MPL
 ti-dəl-j-om
 SM.3FSG-want.IPFV-OM.3MPL
 Intended: ‘The teacher wants the students to be reading.’

This could be seen as an indication that that *ki*-clause complements are tense defective, similar to Greek.

However, while complement *ki*-clauses do not host tense auxiliaries, they do permit for temporal adverbials that refer to a time that is different from the the time of evaluation for the matrix clause.

- (30) a. [ʔit-a səbajti ts’ibāh ki-t-xəjjid]
 that-FSG woman tomorrow IRR-SM.3FSG-leave.IPFV
 ji-gibaʔ-a nəjr-u
 SM.3MSG-need.IPFV-OM.3FSG AUX.PST-SM.3MSG
 ‘The woman needed to leave tomorrow.’
 b. ʔit-a məmhīr n=ət-om təmhāro [t₁ ts’ibāh
 that-FSG teacher ACC=that-MPL student tomorrow
 ki-xejd-u] delij-at-om
 IRR-leave.IPFV-SM.3MPL want.GER-SM.3FSG-OM.3MPL
 nəjr-a
 AUX.PST-SM.3FSG
 ‘The teacher wanted the students to leave tomorrow.’

Importantly, we find nominal hyperactivity in both constructions alongside this evidence for semantically expressed tense in the embedded clause. This suggests that hyperactivity in *ki*-clause complements does not reflect an absence of semantic tense.

Given the facts in (29), one might instead attribute the defectiveness of the embedded clause to the lack of morphosyntactic tense. The nominal hyperactivity observed in the context of *kəmzi*-clauses, so called for the complementizer *kəmzi* on the highest verbal element of the clause (see Spadine 2020; Cacchioli in preparation), speaks against this alternative as well.

- (31) *ʔit-om səbat* [*ʔit-a səbajti ki-t-xəjjid*
 that-MPL people that-FSG woman IRR-SM.3FSG-leave.IPFV
kəmzi-nəbər-a] *rafi-om-wa*
 COMP-AUX.PST-SM.3FSG forget.GER-SM.3MPL-OM.3FSG
 ‘Those people forgot that the woman would leave.’

The embedded argument *ʔita səbajti* ‘the woman’ in the example above controls subject agreement on the embedded predicate as well as object agreement on the matrix predicate. This is so even in the presence of a tense auxiliary *nəbərə* in the embedded clause. The morphosyntactic deficiencies of *ki*-clauses, therefore, are not obviously responsible for creating an environment that permits hyperactivity.⁹

In sum, a theory of defectiveness that is built on the concept of semantic or morphosyntactic tense does not appear to be suited to provide an explanatory account for patterns of hyperactivity in Tigrinya. With respect to the semantic expression of tense, all of the environments investigated above have some expression of this feature and all host hyperactive nominals. The data above also demonstrate that hyperactivity can be observed both in the presence and absence of morphosyntactic tense, making this an unlikely factor as well. Moreover, the constancy of the expression of tense in *ki*-clauses makes it an unlikely factor in explaining when we observe hyperagreement or hyperraising in Tigrinya.

Finally, it would seem that the expression of tense also cannot be used as a predictive factor for hyperactivity cross-linguistically. In Greek we observed hyperactivity in the absence of semantic tense while its presence permits hyperactivity in Tigrinya. We also observe hyperactivity in both the presence and absence of morphosyntactic tense in Tigrinya. This notably contrasts with a language like English, in which hyperactivity is never observed in the presence of morphosyntactic tense. These results support previously reached conclusions against theories of hyperactivity that rely on the semantic and morphosyntactic expression of tense (see also Zeller 2006 on Zulu).¹⁰

⁹While space precludes a full discussion of *kəmzi*-clauses, available evidence points towards a long-distance hyperagreement analysis of the type proposed for pseudo-modal predicates in §3.1

¹⁰Boran Kim (pers. comm.) notes that aspect could potentially serve as a defective category in Tigrinya. The fact that *ki*-clause complements only appear with imperfective verb forms could be taken to motivate the hypothesis that imperfective aspect is somehow defective. While present space precludes a full demonstration, hyperactive behaviors are observed in the context of any verb form. This suggests that aspect is also an unlikely candidate for providing an

5 Argument structure alternations predict hyperactivity

The preceding discussion leads us to the question of what actually is responsible for the patterns of hyperactivity in Tigrinya and cross-linguistically. I will suggest that hyperactivity in Tigrinya is motivated, at least in part, by the formal requirements of verbal functional heads in the superordinate clause. This proposal forms part of an emerging literature that has similarly concluded that constraints on A-relations do not necessarily reflect factors involved in the licensing of nominal constituents, but instead more generally reflect the requirements of heads on the verbal spine (Zyman 2018; Halpert 2019; Fong 2019; Lee & Yip 2024; Halpert & Zeijlstra 2024).

The crucial observation is that hyperactivity patterns in Tigrinya appear to be predictable on the basis of the argument structure of the matrix predicate. The intensional predicate /dlj/ ‘want’ is a transitive predicate, as evidenced by the ability to undergo passivization. However, as Cacchioli & Overfelt (in preparation) observe, the *ki*-clause selecting pseudo-modal /gbʔ/ ‘need’ shows several indicators of being an unaccusative predicate.

An initial piece of evidence to this effect, which was originally observed by Tesfay (2016), is found in the morphological form of the predicates. The imperfective form of the verb /gbʔ/, which is shown in isolation in (32), has the templatic shape CiC(C)aC. This is the expected form of intransitive members of its verb class. This contrasts with the transitive /dlj/ ‘want’, which shares the templatic shape CäC(C)iC with the paradigmatic transitive verb /sbr/ ‘break’ from §2.2.

(32) *Imperfective of /gbʔ/*

ji-gibbaʔ

SM.3MSG-need.IPFV

(33) *Imperfective of /dlj/*

ji-dəllij

SM.3MSG-want.IPFV

The unaccusative argument structure of /gbʔ/ is also reflected in the obligatorily default 3MSG subject agreement that was identified in §3.1. It is not possible to interpret these structures with a referential subject, nor is it possible to overtly provide one in the way that is attempted with *nissu* ‘he’ in (34):

explanatory theory of the hyperactivity facts in Tigrinya.

- (34) (***nissu**) [?it-a səbajti ki-t-xəjjid]
 3MSG.NOM that-FSG woman IRR-SM.3FSG-leave.IPFV
 ji-giba?-a
 SM.3MSG-need.IPFV-OM.3FSG
 Intended: ‘He needs the woman to leave.’

These are properties that partially follow from the unaccusativity of /gb?/. The lack of an external theta-role means that /gb?/ cannot introduce a referential argument for the grammatical subject position without incurring a violation of the Θ -Criterion (Chomsky 1981).

The default subject agreement that appears on the pseudo-modal verb requires slightly more discussion. As the example in (35) shows, the logical subject ?ita səbajti ‘the woman’ of a pseudo-modal construction cannot control matrix subject marking.

- (35) * ?it-a səbajti [t₁ ki-t-xəjjid]
 that-FSG woman.F IRR-SM.3FSG-leave.IPFV
 ti-gibba?-(a)
 SM.3FSG-need.IPFV-OM.3FSG
 ‘The teacher wants the students to read the book.’

This reveals that these constructions permit only expletive subjects in the matrix, meaning that the Raising-to-Subject derivation that is represented in (35) must be unavailable. Instead, as argued in §3.1, hyperagreement in pseudo-modal constructions necessarily arises via long-distance hyperagreement.

This fact led Overfelt & Cacchioli (under review) to speculate that there is no Raising in Tigrinya. However, accepting the argument that the intensional predicate /dlj/ ‘want’ of §3.2 permits Raising-to-Object derivations, this claim would be too strong. It would even be too strong to claim that there is a prohibition specifically against Raising-to-Subject in Tigrinya. We have observed a Raising-to-Subject derivation in the context of a passivized intensional predicate in the repeated example below:

- (36) ?it-om təmharo₁ b=it-i məmħir [t₁
 that-MPL student.PL INS=that-MSG teacher
 ni=ki-xəjd-u] tə-dəlij-om
 ACC=IRR-leave.IPFV-SM.3MPL INTR-want.GER-SM.3MPL
 ‘The students are wanted by the teacher to leave.’

Putting together the conclusions that have been reached here and in the previous sections, it is possible to characterize the empirical landscape for the hyperactivity patterns that we observe with complement *ki*-clauses in Tigrinya as follows:

- (37) *The hyperactivity patterns of ki-clauses in Tigrinya*
- a. Transitive predicates show Hypreraising-to-Object.
 - b. Passivized transitive predicates show Hypreraising-to-Subject.
 - c. Unaccusative predicates show Hyperagreement.

In short, the hyperactive behavior of nominal constituents embedded in *ki*-clause complements appears to be predictable on the basis of the argument structure of the embedding predicate.

This is a way of characterizing the hyperactivity patterns in Tigrinya that modern theories of argument structure alternations are well-suited to handle. In particular, we can start from the assumption that an inventory of functional verbalizing v^0 heads, or possibly a series of such heads, are responsible for syntactically encoding the argument structure properties of the predicate (Kratzer 1996; Chomsky 2000; Pyllkkänen 2008; Alexiadou et al. 2015 and numerous others).

This makes it possible to propose that v^0 heads in Tigrinya vary on the basis of their specification for at least two different types of morphosyntactic features to deliver the inventory in Table 3.

Table 3: An inventory of verbal functional heads and the effect on derivations in Tigrinya

v_{TRAN}^0	[EPP , ϕ]	Hyperraising-to-Object
v_{PASS}^0	[EPP]	Hyperraising-to-Subject
v_{UNAC}^0	[ϕ]	Hyperagreement

Along the lines of the analysis of hyperraising proposed by Zyman (2018), one of these features is a movement-licensing feature, such as an EPP feature in any of its modern instantiations. This feature could be seen as the motivating factor for shifting arguments into a derived object position and, thereby, driving Raising derivations for transitive and passivized predicates. The second type of feature, which reflects fairly standard beliefs about object agreement, is a set of ϕ -features (e.g., Baker & Kramer 2018; van der Wal 2018). The presence of this

set of features determines that agreement accompanies movement in the case of a transitive predicate or that an unaccusative predicate simply agrees with an embedded argument.

Available space precludes providing an articulated model of precisely how this kind of variation would deliver the intended results. However, the hope is that these suggestions might lay the groundwork for future exploration.

6 Conclusion

This paper has investigated patterns of nominal hyperactivity in the context of *ki*-clause complements in Tigrinya. I have argued that these patterns are neither explained nor predicted by analyses that rely on concepts such as nominal-licensing and nominal deactivation. In short, nominals remain active for additional A-relations despite the fact that they are Case-licensed in domains that otherwise are not defective in any obvious sense.

Instead, it was found that the relevant feature for predicting the nature of hyperactivity patterns in Tigrinya was the argument structure of an embedding predicate. This motivated the idea that nominal hyperactivity may best be explained by variation in the formal properties of verbal functional heads (see also Zyman 2018; Halpert 2019; Fong 2019; Lohninger et al. 2022; Lee & Yip 2024; Halpert & Zeijlstra 2024).

Beyond fully articulating this idea and demonstrating its efficacy for Tigrinya, there are numerous open issues that remain for future research. Most significantly, perhaps, is that it remains to be demonstrated that variation on the dimension of argument structure offers an account for (non-)hyperactivity cross-linguistically. Additionally, even within Tigrinya, there remains the question of how clause-size interacts with hyperactivity pattern seeing as *kəmzi*-clause appear to be opaque for Raising. The hope is that such issues might be reduced to the properties of embedded clauses and the verbal functional heads that are trying to access their contents.

Abbreviations

1 = first person, 2 = second person, 3 = third person, ACC = accusative, AUX = auxiliary verb, COMP = complementizer, F = feminine, GER = gerundive, INS = instrumental, INTR = intransitive, IPFV = imperfective, IRR = irrealis, LOC = locative, M = masculine,

NOM = nominative, NPST = non-past, OM = object marker, PL = plural, POSS = possessive, PRF = perfect, PST = past, SBJV = subjunctive, SG = singular, SM = subject marker.

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